```
!pip install gensim numpy matplotlib

import gensim
import numpy as np
import matplotlib.pyplot as plt
```

```
Collecting gensim
  Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (8.4 kB)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim
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Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1-
Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.9 MB)
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Installing collected packages: gensim
Successfully installed gensim-4.4.0
```

```
import gensim.downloader as api

model = api.load("glove-wiki-gigaword-100")
```

```
[==================================================] 100.0% 128.1/128.1MB downloaded
```

```
# Target words
X = ["man", "male", "boy", "brother", "he", "him"]
Y = ["woman", "female", "girl", "sister", "she", "her"]

# Attribute words
A = ["career", "business", "office", "salary", "professional"]
B = ["family", "home", "children", "marriage", "parents"]
```

```
def cosine_similarity(w1, w2):
    return np.dot(w1, w2) / (np.linalg.norm(w1) * np.linalg.norm(w2))
```

```
def association(w, A, B):
    return np.mean([cosine_similarity(model[w], model[a]) for a in A]) - \
           np.mean([cosine_similarity(model[w], model[b]) for b in B])
```

```
def weat_score(X, Y, A, B):
    return np.sum([association(x, A, B) for x in X]) - \
           np.sum([association(y, A, B) for y in Y])
```

```
def effect_size(X, Y, A, B):
    combined = X + Y
    assoc_values = [association(w, A, B) for w in combined]

    mean_x = np.mean([association(x, A, B) for x in X])
    mean_y = np.mean([association(y, A, B) for y in Y])
```

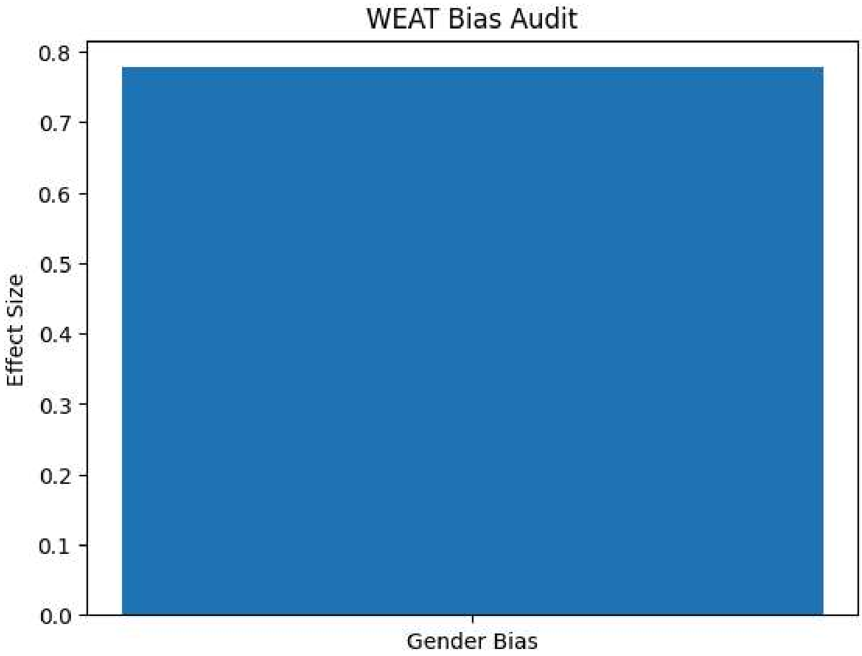
```
    return (mean_x - mean_y) / np.std(assoc_values)
```

```
score = weat_score(X, Y, A, B)
es = effect_size(X, Y, A, B)

print("WEAT Score:", score)
print("Effect Size:", es)
```

```
WEAT Score: 0.4135036
Effect Size: 0.7773372
```

```
plt.bar(["Gender Bias"], [es])
plt.ylabel("Effect Size")
plt.title("WEAT Bias Audit")
plt.show()
```



Start coding or generate with AI.